

Project Title

Deadline Guardian

Tagline

A Chrome extension that turns Canvas deadlines into active browser reminders.

1. Problem Statement

Many students rely on Canvas to track assignments, quizzes, and project deadlines. However, the Canvas to-do list can be easy to ignore because students still need to open Canvas and check it manually.

As a result, students may forget upcoming tasks, miss deadlines, or only remember assignments when it is already too late.

2. Proposed Solution

Deadline Guardian is a Google Chrome extension that connects to the **Canvas API** and fetches a student's upcoming assignments.

Instead of only showing deadlines inside Canvas, the extension displays them directly in the browser and sends notifications when a deadline is near.

The goal is to make deadlines more visible, active, and harder to miss.

3. One-Sentence Pitch

Deadline Guardian helps students avoid missed Canvas deadlines by showing upcoming assignments in a Chrome extension and sending browser alerts before they are due.

4. Target Users

The target users are:

- students who use Canvas for classes
 - students who often forget assignment deadlines
 - students managing multiple courses
 - students who do not regularly check the Canvas to-do list
 - student groups working on projects with several deadlines
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5. Key Features

Feature	Description
Canvas Connection	API User enters their Canvas URL and API token to connect the extension to Canvas.
Deadline Sync	The extension fetches active courses and upcoming assignments from Canvas.
Upcoming Deadline List	Displays assignments sorted by nearest due date.
Countdown Timer	Shows how much time is left before each deadline.
Urgency Labels	Labels tasks as Normal, Soon, Urgent, or Overdue.
Browser Notifications	Sends alerts when a deadline is close.
Open in Canvas	Lets users open the assignment directly from the extension.

6. How It Works

1. The student installs the Chrome extension.
2. The student opens the extension popup.
3. The student enters their Canvas URL and API token.
4. The student clicks "Sync Deadlines."
5. The extension requests active courses from Canvas.

6. The extension requests assignments from each course.
 7. The extension filters assignments with due dates.
 8. The extension sorts assignments by nearest deadline.
 9. The extension displays the deadlines in the popup.
 10. The extension sends browser notifications when deadlines are near.
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7. Example User Scenario

A student has a task on Canvas:

Assignment: Chapter 4 Draft

Course: Capstone

Due Date: May 8, 2026 at 11:59 PM

Deadline Guardian fetches this assignment and shows:

Chapter 4 Draft

Course: Capstone

Due in 2 days

Status: Soon

When the deadline is close, the student receives a browser notification:

Deadline Reminder:

Chapter 4 Draft is due tomorrow.

Later, the extension can send another alert:

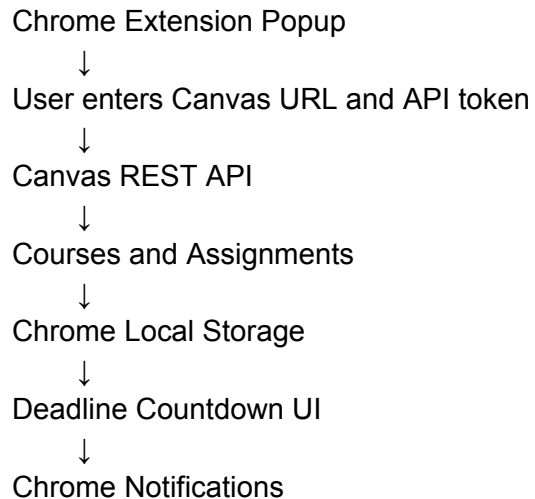
Urgent Reminder:

Chapter 4 Draft is due in 3 hours.

8. Technology Stack

Part	Technology
Platform	Google Chrome Extension
Frontend	HTML, CSS, JavaScript
API	Canvas REST API
Storage	Chrome Local Storage
Notifications	Chrome Notifications API
Background Tasks	Chrome Alarms API
Authentication for MVP	Canvas Personal Access Token

9. System Architecture



10. MVP Scope

For the hackathon version, the project will focus on the simplest working version.

Must-Have Features

Feature	Status
Extension popup interface	Required
Save Canvas URL and token	Required
Fetch Canvas courses	Required
Fetch assignments with due dates	Required
Display upcoming deadlines	Required
Sort deadlines by date	Required
Show countdown	Required
Send browser notification	Required

Optional Features

Feature	Priority
Manual deadline entry	Medium
Multiple notification levels	Medium
Mark task as completed	Medium
Dark mode	Low
AI priority suggestions	Low
OAuth login	Future version

11. Why This Project Matters

Canvas already stores assignment deadlines, but it does not always make them noticeable enough. Many students only remember to check deadlines when they open Canvas.

Deadline Guardian improves this by bringing deadlines directly into the browser and reminding students before the deadline becomes urgent.

This helps students:

- avoid missed submissions
 - manage multiple courses better
 - reduce last-minute cramming
 - stay aware of upcoming academic tasks
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12. Competitive Advantage

Unlike a normal to-do list, Deadline Guardian does not require students to manually create every task.

Unlike the Canvas to-do list, Deadline Guardian is not passive.

It combines Canvas data with browser-based alerts, making the reminders more active and accessible.

13. Hackathon Feasibility

This project is realistic for a **3–8 hour hackathon** because it does not require a backend, database, or full login system.

The MVP only needs:

manifest.json
popup.html
popup.css
popup.js
background.js

The extension can store settings and assignments locally using Chrome storage.

14. Demo Plan

During the demo, the presenter can show:

1. Opening the Chrome extension popup.
 2. Entering a Canvas URL and API token.
 3. Clicking "Sync Deadlines."
 4. Displaying upcoming Canvas assignments.
 5. Showing countdowns and urgency labels.
 6. Triggering or explaining a browser notification.
 7. Clicking an assignment to open it in Canvas.
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15. Final Pitch

Deadline Guardian is a Chrome extension that helps students stay ahead of Canvas deadlines. It connects to the Canvas API, fetches upcoming assignments, displays them in a clean countdown list, and sends browser notifications before tasks are due. Instead of relying on students to constantly check Canvas, Deadline Guardian makes deadlines active, visible, and harder to miss.